MOULOUD SERIR

Full-Stack and AI Developer | React, Next.js, TypeScript, Node.js, Python

Sherbrooke, QC (Montréal hybrid or remote) • mouloudserir24@gmail.com • Available from January 2027
Portfolio: mouloud08.github.io • github.com/Mouloud08 • linkedin.com/in/mouloud-serir-0a4842257

SUMMARY

Bilingual web developer and CS student graduating in Dec 2026, with two AI and full-stack internships and one web internship. I build web apps end to end with React, Next.js, TypeScript and Node.js, and AI features (agents, RAG, tool calls).

PROFESSIONAL EXPERIENCE

Intelligence Industrielle — Montréal, QC

Software Engineering Intern (second internship) — May 2026 – Aug 2026

TypeScript | React | Next.js | TanStack Query | Node.js | MongoDB | BigQuery | n8n | Playwright

- Built an internal app where **AI agents orchestrated in n8n** create versioned BI reports from live BigQuery data, edited via chat.
- **Redesigned the UI/UX** of an internal project lifecycle app with my supervisor and product manager, built its layouts and coded its multi-agent router, tool calls and **two real-time voice agents** (one on Gemini Live).
- Modernized an existing internal user management app (admin gateway): UI/UX, complementary features, simpler flows.
- Made long-running AI jobs reliable: async processing, persisted state, retries, JSON Schema validation of LLM outputs.
- Improved **front-end performance** with TanStack Query caching, IndexedDB and pagination, profiled with Chrome DevTools.
- Instrumented products with Google Analytics, with telemetry fed into an internal observability tool.
- Prototyped apps in single-file HTML for demos, then rebuilt them in Next.js or Vite with AI coding agents, TDD and E2E tests.

Software Engineering Intern (first internship) — Sep 2025 – Dec 2025

TypeScript | React | Next.js | IndexedDB | MongoDB | n8n | Airtable | Playwright

- Built full-stack, with another intern, a **production client app** for technical drawings: two AI agents extract the data, a third compares them with its own and keeps the best value per field (**confidence score**); the app then computes material and cost.
- Built v1 of an internal project lifecycle tool with LLM agents (n8n, Airtable), **adopted internally across company products**.
- Built an AI image generation and editing app on a **node-based canvas** where users chain steps (generation, image fusion, edits).

Université de Sherbrooke — Sherbrooke, QC

Research Assistant – Spatio-temporal Forecasting — Sep 2026 – Present

Supervisor: Prof. Djemel Ziou | Python | ST-LLM+

- Forecasting emergency call volumes per county (MRC) and 8-hour shift with a spatio-temporal LLM.

Teaching Assistant – IFT320 Operating Systems — Jun 2026 – Aug 2026

- Graded C/Linux assignments for **38 students**, with feedback on synchronization, concurrency and code quality.

Research Assistant – Application Development — Feb 2026 – May 2026

Supervisor: Prof. Djemel Ziou | React | Vite | Three.js | D3.js | WebXR

- Built a curriculum explorer: **interactive 3D graph of courses**, grid views and an AI assistant restricted to controlled actions.
- Built an ETL pipeline for a CCSLL application in March 2026 (consulting contract).

Borealis — Magog, QC • Remote

Web Developer Intern — Jan 2025 – Apr 2025

JavaScript | SQL | XML | Jira

- Delivered client-requested features from Jira tickets; traced and fixed a recurring interface crash with SQL/XML patches.

PROJECTS

Nexora, AI-Augmented Learning Platform – Capstone Project (team lead) — Sep 2026 – Present

Supervisor: Prof. Nadia Tahiri • IFT596, team of 5 • my areas: full-stack, AI/RAG, UI/UX • in development

- Leading a team of 5: a RAG tutor that cites its sources, a dashboard of misunderstood concepts and a course-content agent.

Shared Infrastructure to Ship Apps Fast – Personal Project — Oct 2026 – Present

- Template repo per app, shared Supabase auth and database with **per-app isolation** (row-level security), Cloudflare Pages.

Atelier, Design System for My Apps – Personal Project — Oct 2026 – Present

- Designed and built with AI coding agents a shared **design system** (React, TypeScript, Tailwind): design tokens, components, 13 page templates and accessibility modes, documented as skills that coding agents follow; used in Nexora and a transit app.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Java, C#, C/C++; French and English
Front-End: React, Next.js, Tailwind CSS, Redux, HTML/CSS, responsive design, accessibility, SEO, A/B tests, CMS
Back-End & Data: Node.js, REST APIs, PostgreSQL, MongoDB, Supabase, BigQuery, microservices
AI & Agents: OpenAI, Claude and Gemini APIs, RAG, embeddings, pgvector, MCP, LangGraph, LangChain
LLM Quality: prompt engineering, LLM evaluation, RAG metrics, agent consensus, cost and token tracking, tracing
Testing & DevOps: TDD, Jest, Vitest, ESLint, CI/CD, GitHub Actions, Docker, Vercel, Agile/Scrum
AI Coding Tools: Claude Code, Codex, GitHub Copilot, Cursor, skill design for coding agents
Soft Skills: autonomy, initiative, communication, teamwork, leadership, creativity, adaptability, problem-solving

EDUCATION

B.Sc. in Computer Science (Co-op), Université de Sherbrooke, Dec 2026 • AWS Certified Solutions Architect (in progress)